Matteo Pignataro

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PROFESSIONAL SUMMARY

Software engineer with hands-on experience in C/C++, embedded systems, real-time software and robotic platforms. Experienced in low-level communications, ROS 2, Docker and hardware-software integration through autonomous racing and experimental rocketry projects. Strong background in low-level development, system integration and multidisciplinary technical coordination.

EDUCATION

Master of Computer Science and Engineering at Politecnico di Milano — *09.2022 - 12.2025*

- Final Mark: 110/110 with honors
- Thesis Title: A Scalable and Bandwidth-Aware Telemetry Framework for Autonomous Racing Vehicles

Bachelor of Computer Science and Engineering at Politecnico di Milano — *09.2019 - 09.2022*

- Final Mark: 105/110

High School Diploma in Computer Science at I.T.I.S. Stanislao Cannizzaro — *09.2014 - 06.2019*

- Final Mark: 100/100 with honors

WORK EXPERIENCE

System Integrator at PoliMOVE Racing Team — *02.2025 - Now*

Pursued a Master's degree in Computer Science and Engineering, with a thesis focused on flexible and bandwidth-aware smart telemetry framework for autonomous racing within the PoliMOVE research group. Experienced frequent international travel as part of the team, including extended technical trips and competitions abroad. On July 24th, the team achieved a historic milestone by securing the first place in the inaugural american road-course autonomous race at Laguna Seca Speedway in California.

Achieved the following goals:
- Developed and integrated multithreaded software within a complex ROS2 project for autonomous vehicles;
- Designed and deployed Docker development environments integrated with Gitlab CI/CD pipelines;
- Designed and developed a fully code generated C++ CAN bus interface from DBC specification files;
- Designed a custom network bandwidth-aware protocol for telemetry;
- Performed code reviews and contributed to merge request discussions to ensure fail-safe software quality and consistency.

Project Manager at Skyward Experimental Rocketry — *09.2023 - 10.2024*

Worked as Project Manager for the Lyra rocket, coordinating a multidisciplinary team of technical leaders in designing and building a paraffin-based sounding rocket. Managed project deadlines and coordinated the work of 150 people to ensure the rocket was competition-ready. On September 14th, the rocket was successfully test-launched in Roccaraso, performing as expected and confirming the design's quality. Served as team leader at the European Rocketry Challenge (EuRoC) in Portugal, where the team secured the first place, winning the competition with a perfectly nominal flight.

Software Team Leader at Skyward Experimental Rocketry *09.2022 - 09.2023*

Worked as Software Team Leader for the Gemini Project. Managed a team of 15 people and worked closely with the electronics department to review hardware specifications and schematics. Studied, developed, and implemented a custom communication protocol between the Ground Station and the Ground Segment Equipment to guarantee collision avoidance for low-rate radio half-duplex communications.

Achieved the following goals:
- Developed an interconnected system of 4 computers via CAN bus communication;
- Debugged low-level issues through the usage of oscilloscopes and logic analyzers;
- Learned the dynamics of the design process as part of the technical team;
- Transformed hybrid rocket engine operational requirements into software and system requirements;
- Led the launch procedures for the hybrid rocket as flight controller.

Software Team Member at Skyward Experimental Rocketry *10.2021 - 09.2022*

Worked as part of the software and control team for the Pyxis Project. Programmed the flight software for the Payload bay computer and worked with the Recovery team during parafoil testing for flight characterization and control algorithm tuning. Have been part of the best team in Europe settling the absolute first place and winning the competition with a nominal flight.

Achieved the following goals:
- Worked with an embedded system developed around the STM32F429 microcontroller;
- Developed real-time software using the Miosix RTOS and performed low-level debugging with GDB;
- Deployed software with standard communication protocols: SPI, UART, I2C, and CAN;
- Developed low-level device drivers from hardware datasheets and reference manuals;

SKILLS

Programming: C, C++, Python, Java, Git

Language: Italian, English

Other: CAD, Electronics, Hand Soldering, Docker

CERTIFICATES

Test of English for International Communication (TOEIC): B2 (955 points) *04.2022*

Cambridge First Certificate (FCE): B1 *05.2018*

INTERESTS

Computer Graphics

One of main interests is computer graphics. Studied and used the Vulkan and OpenGL APIs and to write a low-level, object-oriented framework for them. Also designed a 3D visualizer for rocket trajectory and attitude to better visualize flights from the ground station.

Electronics

Passionate of analog and digital electronics. At university studied the fundamentals of electronics and completed a course in digital electronics design. Additionally, completed Radio amateur certification. Presented twice at the Rome Maker Faire showcasing two high-voltage projects. Through these experiences, learned the usage of essential electronics instrumentation such as oscilloscopes, logic analyzers, and function generators, which enabled me to design and build analog amplifiers and multiple solid-state Tesla coils at home.